Epistemic Generation as Mechanization Placement
A Corpus Study of Research-Move Vocabularies and Evaluation Boundaries

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Abstract

Research systems need to propose useful next moves before final claims exist. This paper studies whether those moves can be named, stress-tested, and partly mechanized while preserving the difference between recognition, action, and judgment. The evidence comes from one operated research corpus and three connected empirical layers. First, a corpus-level operationalization of Gowers's theory-builder/problem-solver distinction finds a symmetric own-corpus advantage: a theory-builder vocabulary covers theory-builder arcs at 58.1% but problem-solver arcs at 20.7%, while a problem-solver vocabulary covers problem-solver arcs at 65.9% but theory-builder arcs at 19.1%. A same-culture random split produces only a 3.0 percentage-point gap. Second, a later eight-subfield remine revises the binary into a broader structural vocabulary: six shared-core operations, eight broadly shared operations, and four peripheral operations over 1,214 moves. That vocabulary transfers beyond its original setting and exposes measurable reliability structure, including a methodological warning from a PDE rescore. Third, prompt and workflow experiments identify the interface that makes the vocabulary operational: passive labels and broad menus carry little causal force; typed evidence contracts, source-bound residual-to-check edges, rejected confusers, and checked action schemas change downstream behavior in replay and synthetic workflow settings. Meta-language experiments add a further result: meta-operator semantics are separable from adjacent lower-tier moves and can sharpen already-parsed evidence artifacts. The resulting thesis is mechanization placement: a research-move vocabulary becomes scientifically useful when it assigns a move to the right role–recognition, human review, evidence contract, deterministic check, policy feature, action schema, or judgment.

1 Introduction

Epistemic verification asks whether a claim is adequately supported. Epistemic generation asks how a research process chooses the next move before such a claim exists. The move might be a reformulation, special case, analogy, falsifier, auxiliary object, decomposition, evidence boundary, or decision to stop a branch.

This paper makes three contributions. First, it gives a corpus-level empirical operationalization of Gowers's theory-builder/problem-solver distinction [Gowers, 2000]. Two curated mathematical styles induce different research-move vocabularies, and the difference survives cross-corpus scoring and a same-culture negative control. The result gives the distinction an empirical measurement surface without forcing all mathematics into two types.

Second, the paper revises the binary into a broader structural vocabulary of research moves. The later vocabulary explains why the original split was real but incomplete: theory-building and problem-solving emphasize different bands inside a wider space of reformulation, abstraction, decomposition, local-to-global assembly, invariance, translation, approximation, and constraint propagation.

Third, the paper identifies where such vocabularies become useful for research systems. The experiments compare passive labels and large menus with typed evidence fields, residual-to-check edges, rejected nearest confusers, and executable action schemas. This is the mechanization-placement thesis. Research-move language becomes operational when it changes what gets checked, blocked, converted into an artifact, or executed next.

For machine learning readers, the paper contributes an evaluation framework for research agents. It treats generation as a sequence of research moves rather than final-answer production, shows that generic evaluation vocabularies can misclassify domain-specific reasoning, and proposes typed intermediate representations for evaluating whether a system selected, checked, and executed the right move.

2 Evidence Base, Reproducibility, and Method

The unit of analysis is a research move: a bounded step that changes a problem representation, admissible object, proof strategy, evidence standard, decomposition, validation frame, or next action. A vocabulary covers a move when a frozen operation gives a mechanistic fit under the scoring rubric.

Coverage is diagnostic. It is not the target being optimized. High coverage can indicate transfer, but it can also indicate force-fitting. The experiments therefore use cross-corpus scoring, negative controls, adversarial raters, decoys, wrong-card controls, source-cluster blocking, and downstream consequence checks. The coverage numbers should therefore be read together with their controls. Raw fit to a vocabulary is weak evidence; fit that survives wrong-card, decoy, source-only, and downstream-consequence checks is the stronger signal.

Table 1 summarizes the evidentiary roles in this paper.

Table 1: Evidence tiers used in the paper.

Evidence tier	Role in the argument
Saved-output corpus results	Main quantitative claims: two-cultures split, negative control, structural-vocabulary coverage, and out-of-distribution transfer.
Cross-family and multi-rater audits	Reliability checks, especially for the PDE correction and catalog confusion analysis.
Replay and synthetic workflow experiments	Mechanism evidence for evidence carriers, residual-to-check edges, boundary cards, and action schemas.
Human and production-trace validation	Next validation layer for expert use and prospective deployment claims.

The core percentages are recomputed from durable records: stored move lists, frozen vocabularies, scoring files, JSON verdicts, and reproducer scripts. Later workflow experiments are treated as mechanism evidence: they show which carrier changes downstream behavior under controlled conditions. The reproducibility packet includes saved-output verifiers for the headline corpus claims, PDE joint-vocabulary scoring artifacts, meta-language score files, and compact workflow replay scorers. The main quantitative claims are tied to recomputable artifacts rather than unreleased model transcripts.

3 A Corpus-Level Two-Cultures Result

The first corpus contains theory-building arcs including Wiles, Grothendieck, Lurie, Scholze, Riemann, Newton, Einstein, and Polya/Hadamard. The resulting theory-builder vocabulary includes moves such as foundational object redefinition, cross-domain unification, parameter-space internalization, vocabulary lifting, strategic specialization, diagonal self-application, and concept revision [Polya, 1945, Lakatos, 1976].

A sister corpus contains problem-solving arcs including Erdos discrepancy, Green–Tao, Hales–Jewett Polymath, Szemeredi regularity, Roth, Behrend, Furstenberg, and Ramsey/Erdos–Szekeres. The resulting problem-solver vocabulary emphasizes partitioning, governed iterative refinement, theorem-use transfer, rank induction, and estimate chaining.

Table 2: Cross-distribution coverage for the two mined vocabularies.

Vocabulary	Theory-builder corpus	Problem-solver corpus
Theory-builder vocabulary	58.1%	20.7%
Problem-solver vocabulary	19.1%	65.9%

The average own-corpus advantage is 42.1 percentage points. The result is symmetric: each vocabulary fits its own corpus far better than the other. A same-culture random split produces only a 3.0 percentage-point gap, far below the curated two-cultures gap. The result therefore gives the Gowers distinction an empirical operational form: two mathematical styles induce different research-move vocabularies, and the gap is not reproduced by a random partition of one style. This is a descriptive operationalization, not a causal identification of the Gowers axis against every possible corpus-construction factor. Subfield, era, and selection effects may contribute to the gap; the claim here is that the curated contrast yields a measurable research-move difference that a same-style random split does not reproduce.

The novelty claim is precise. We are not aware of prior work giving a corpus-level empirical operationalization of Gowers's theory-builder/problem- solver distinction. Prior work has studied mathematical practice, proof reading, examples, collaboration, consensus, and subfield prestige [Cranshaw and Kittur, 2011, Gargiulo et al., 2022, Weber and Mejia-Ramos, 2014, Lockwood et al., 2016, Mejia-Ramos et al., 2021, Wagner, 2022, Schlenker, 2020]. Those studies are adjacent rather than competing: they do not directly test whether theory-building and problem-solving styles differ by cross-coverage of research-move vocabularies.

4 From Two Cultures to a Structural Vocabulary

The two-cultures result was a starting point, not the final taxonomy. A subsequent eight-subfield analysis covered 1,214 moves across 64 mathematical arcs and compressed them into six shared-core operations, eight broadly shared operations, and four peripheral operations.

This revision generalizes the original result. Theory-builder and problem-solver mathematics emphasize different bands inside a larger structural language. Some moves are shared, some are subfield idioms, and some apparent culture-specific moves become aliases after the wider remine.

Table 3: Layered structural vocabulary from the eight-subfield remine.

Tier	Operations
Shared core	Problem Reformulation and Reduction; Generalization and Abstraction; Decomposition and Recomposition; Local-to-Global Assembly; Canonical Form and Invariance; Cross-Domain Translation.
Broadly shared	Iterative Refinement; Recursive Decomposition; Duality and Adversarial Framing; Layered Approximation and Convergence; Extremal Method; Probabilistic and Stochastic Methods; Dimensional and Structural Lifting; Constraint Imposition and Propagation.
Peripheral	Characterization by Obstruction; Internalization and Self-Reference; Axiomatization and Foundational Repair; Controlled Universe Extension.

The vocabulary also transfers beyond the original mathematical corpus. A business held-out set reaches 57.9% shared-plus-broad coverage. Four sparse 2026 specialist papers reach 75.2%. These results support measured portability, not unrestricted universality: scope, residuals, and confusers are recorded rather than absorbed by definition.

5 Methodological Warning from the PDE Rescore

The PDE rescore is a methodological result for the evaluation of frontier agents and research systems. Domain-specific reasoning can look absent when an evaluation vocabulary omits the domain-specific operations that carry the reasoning. A 12.5% adversarial score on a post-cutoff PDE paper initially looked like a sharp capability boundary for the structural vocabulary. The audit showed a different mechanism: the score came from a single vocabulary, a single adversarial rater, and one difficult PDE paper. When the same paper was rescored against the joint structural vocabulary plus the PDE estimate-craft vocabulary, four raters across two model families covered 95.8–100% of moves under the covered-vs-none metric. Across the five-paper corpus, two-family coverage reached 116/118 moves, or 98.3%, with a 0% decoy match rate.

Table 4: Methodological interpretation of the PDE stress case.

Finding	Current interpretation
Old 12.5% PDE score	Historical single-vocabulary adversarial baseline.
Joint vocabulary PDE audit	95.8–100% covered-vs-none coverage across four raters.
Corpus-wide joint audit	116/118 moves covered by both model families.
Strict full-only coverage	Calibration-sensitive secondary diagnostic.
Remaining uncertainty	Fresh blind PDE corpus, human or expert audit, and operation-confusion analysis.

The warning for machine-learning evaluation is direct. An evaluation that tests frontier agents on domain-specific reasoning while using only generic reasoning categories can manufacture artificial capability boundaries. The expanded language plus PDE estimate-craft receipts covered the former stress case under multi-rater, decoy-controlled scoring. The paper therefore reports covered-vs-none and operation-identity reliability, with strict full-only coverage as a calibration-sensitive secondary metric.

6 Carrier Selection in Agent-Facing Tests

The early agent-facing experiments tested research-move vocabulary supplied as prompt text, feature prose, or a broad catalogue. Those carriers had little effect on the tested outcomes.

Table 5: Carrier-selection evidence from agent-facing tests.

Test	Result	Interpretation
Exact-answer benchmark with primitive prompts	Baseline, static catalogue, one-primitive, and diagnose-then-solve arms each scored 11/26; placebo diagnose-then-solve scored 8/26.	No exact-answer lift from primitive prompt text.
Blind route-choice probes	Generic reasoning won 7/10 comparisons; passive real features, forced real features, and forced placebo each won 6/10.	Primitive text changed rationale vocabulary more than route choice.
Specialized-card transfer test	Schema and swap controls matched or outperformed the target card; the target card beat all controls in 0/24 cases.	The card's added content did not separate from general schema discipline.
Downstream execution test	Ordinary routes received 51 votes; schema routes 47; primitive-feature routes 28.	Execution did not rescue primitive-feature routes.
Consequence-ranking test	Target card scored 4.25/5 vs generic 4.04/5 and beat the strongest control in 2/20 source clusters.	Small score gains did not imply reliable consequence improvements.

The catalogue experiments locate the mechanism more precisely. In one comparison, an action-oriented catalogue improved slightly over a descriptive catalogue, but not over source-only reasoning: source-only scored 6.50/10, the catalogue summary 6.00/10, and the action-oriented catalogue 6.50/10. In a single-card comparison, a correct action card scored 8.875/10 while a plausible wrong card scored 4.375/10. Card content matters when the correct move has already been identified. The central problem is routed selection with disambiguators, nearest-confuser rejection, and evidence artifacts.

7 Implications for Research-Agent Evaluation

The experiments suggest three design requirements for evaluating research agents on open-ended scientific or mathematical work.

First, final-answer accuracy is too coarse for epistemic generation. A research agent may fail by selecting the wrong object, paying the wrong evidence debt, over-updating from a narrow receipt, omitting a necessary falsifier, or continuing after a branch should stop. These failures are often invisible in a single final-answer score.

Second, the evaluation vocabulary must include the domain operations that carry the reasoning. The PDE rescore shows how a generic structural vocabulary can understate capability when estimate-craft moves are absent from the scoring language. The same principle applies beyond PDE: clinical, legal, institutional, and workflow settings each have admissibility and authority conditions that generic reasoning labels may miss.

Third, useful evaluation artifacts should preserve action-relevant state. The most informative records in these experiments are not labels alone. They are typed fields: selected residual edge, rejected nearest confuser, source evidence, satisfied and unsatisfied evidence obligations, action schema, step index, invariant check, and later outcome. These fields make it possible to evaluate whether a system executed the correct research move rather than producing a plausible description of that move.

8 Meta-Language

The meta-language track asks whether some moves operate one level above the base structural vocabulary: changing admissible criteria, making an equivalence class primary, promoting an auxiliary object into the main object of study, or changing the question frame itself.

Table 6: Summary of the meta-language evidence.

Experiment family	Result	Interpretation
Family separation	Correct meta cards beat adjacent lower-tier forced fits; generic and meta-frame-only variants reached ceiling.	Meta families are separable, but easy cases do not test downstream use.
Target-card control	Target cards beat wrong cards and adjacent lower-tier cards; typed-frame-only scored 11/11.	Typed parsing can identify the family; cards control confusers.
Artifact revision	Target-card revision improved frozen typed-frame artifacts from 6.91/8 to 7.91/8 and beat generic revision.	Meta cards sharpen an already-parsed artifact.
Naturalistic transfer	Typed-frame and target-card variants reached ceiling; adjacent and wrong cards failed.	Separability holds once typed roles are visible.
Packaging ablation	Correct content beat wrong content; changing the package alone had little effect.	Content mattered more than syntax on this endpoint.

The supported conclusion is that compact meta-level semantics improve the quality of already-parsed evidence artifacts and distinguish frame-changing moves from adjacent lower-tier moves. Wrong meta cards can mis-shape downstream artifacts, which means the meta layer has causal content. The evidence supports three candidate meta families and argues against premature expansion to a fourth.

9 From Vocabulary to Action Schemas

The strongest later evidence comes from experiments where the vocabulary is compiled into an action schema rather than displayed as a label.

Boundary-card experiments show the pattern. Agents can act correctly from a fully specified boundary card that states the satisfied evidence obligation, unsatisfied evidence obligation, permitted update, blocked update, next-action rule, and false-reading confuser. Raw model extraction of that card is unreliable. Model-only validation improves but still accepts unsafe cards. Rule-backed source-cue validation and action-schema fields recover correct behavior on the tested packets.

The process-control experiments show the same architecture. A compiled action schema with selected residual edge, rejected nearest confuser, source evidence, step index, and required next action reached 1.0 action accuracy in the held-out two-step packet. Label-only controls failed. Free-form automatic compilation failed. Typed class selection helped but was unsafe until source-cue checks and deterministic lowering were added. Open-set refusal was needed to avoid forcing outside cases into known classes.

The resulting architecture is a research-agent contract:

$$\text{source facts} \rightarrow \text{residual/evidence carrier} \rightarrow \text{nearest confuser}$$
$$\rightarrow \text{action program} \rightarrow \text{deterministic check} \rightarrow \text{outcome trace.}$$

This is adjacent to algorithm selection, metareasoning, selective prediction, mixture-of-experts routing, prompt-programming systems, and workflow-agent evaluation [Rice, 1976, Xu et al., 2008, Jacobs et al., 1991, Russell and Wefald, 1991, Geifman and El-Yaniv, 2017, Chen et al., 2023, Ong et al., 2024, Hu et al., 2024, Shi et al., 2025, Nandi et al., 2025, Xu et al., 2024]. The contribution here is narrower: a runtime contract for research agents that binds local source facts to the next inspectable action.

10 Mechanization Placement

The common result across the math corpus, structural vocabulary, agent-prompt tests, meta-language probes, and action-schema experiments is placement. A recurring research move can occupy different roles.

Table 7: Placement roles for research-move vocabulary.

Placement	Appropriate use	Failure mode
Recognition language	Naming and comparing research moves.	Mistaken for a recipe.
Human review aid	Helping reviewers remember obligations and confusers.	Measured by self-reported usefulness instead of blinded decisions.
Evidence contract	Forcing an output into inspectable fields.	Shape is imitated without source alignment.
Deterministic check	Enforcing a crisp pass/fail condition.	Check overreaches beyond its contract.
Policy feature	Supplying state to a decision rule.	Retrospective labels mistaken for prospective policy quality.
Action schema	Binding source facts to next actions and stop rules.	Unsafe if parsed free-form or without refusal.
Judgment	Handling semantic frontier choices.	Hidden behind premature automation.

The practical lesson is to place each move at the strongest level its evidence supports. Structural vocabulary is the recognition layer. Typed receipts and nearest-confuser fields are the evidence-contract layer. Deterministic gates are justified when the contract is crisp. Human judgment remains necessary where the move is semantic, frontier-dependent, or mathematically substantive.

11 Future Validation

The present evidence supports five bounded claims: a two-cultures operationalization, a field-portable structural vocabulary, a corrected PDE audit, a scoped meta-language artifact result, and a carrier-selection account of mechanization. Additional experiments would strengthen external validity and prospective-use claims.

The highest-value follow-on experiments are: (i) a blinded human or expert audit of structural-vocabulary assignments and checklist usefulness; (ii) production or historical-trace replay testing whether compiled contract fields reduce downstream cost, missed obligations, false stops, or wrong actions relative to strong context baselines; (iii) a clean meta-language downstream-consequence test with generic headroom; and (iv) a fresh blind PDE or broader scientific corpus audit using covered-vs-none coverage, operation-identity reliability, decoys, and human review.

12 Broader Impact

The main positive impact is better evaluation discipline for research agents. The paper argues for measuring intermediate research moves, evidence payments, confuser rejection, and action schemas rather than relying only on final-answer scores or broad reasoning labels. This can reduce misleading claims about agent capability and make domain-specific limitations more inspectable.

The main risk is misuse of the vocabulary as an authority signal. A structural label can make a weak research step appear more systematic than it is. The mitigation is built into the placement thesis: labels are advisory unless they are tied to source evidence, falsifiers, deterministic checks, or downstream outcomes. Domain-specific evaluation should report residuals and confusers instead of forcing every move into the nearest available category.

13 Limitations

This paper studies one research system and curated corpora. Its contribution is a measured vocabulary and mechanization theory for research moves.

Most coverage judgments are model-mediated. Cross-family and multi-rater audits strengthen several claims; independent human and expert audits are the natural next validation layer. The paper therefore treats human audit as a future external-validity test, not as completed evidence.

Several agent-facing experiments are small-N mechanism tests. Their point estimates are reported as diagnostics, not as population estimates. Small differences should not be read inferentially; the main evidentiary weight falls on large contrasts, negative controls, decoys, wrong-card tests, and cases where a carrier change alters the downstream artifact.

Coverage measures structural recognition. Generative competence, frontier selection, and mathematical correctness require separate tests.

The evidence supports a compact central vocabulary plus field-specific receipts and confuser guards. It also identifies overlap, low-use operations, and calibration-sensitive full/partial distinctions in some catalog versions.

The meta-language evidence supports separability and artifact refinement under typed conditions. Incremental routing and downstream consequence tests are the next extension.

The action-schema positives are strongest in replay, synthetic, and rule-checked settings. Production-trace tests are the next external-validity step.

14 Conclusion

The paper began with a Gowers-style question: do theory-building and problem-solving arcs induce different research-move vocabularies? In this corpus, they do. The broader lesson is that the binary sits inside a structural vocabulary whose usefulness depends on placement.

Passive labels have little causal force. Correct compact schemas can help after selection. Meta-language can sharpen typed artifacts. Checked action schemas can change downstream behavior. The general lesson is to name recurring research moves, test their boundaries, and mechanize the parts whose contracts can be made inspectable.

A Evidence Map

Table 8: Principal local artifacts behind the paper claims.

Claim family	Saved artifact or source
Two-cultures cross-coverage	`evidence/gp216_queries/gp216_cross_distribution_full.json`
Headline verifier	`evidence/reproducers/verify_gp216_claims.py`
Same-culture negative control	`evidence/gp216_queries/gp216_negative_control.json`
Four-operation compression	`evidence/gp216_queries/gp216_4op_compression.json`
Eight-subfield structural vocabulary	`evidence/gp216_queries/gp216_pathA_8sf_cluster.json`
Business held-out transfer	`evidence/gp216_queries/gp216_business_held_out_OOD.json`
Sparse 2026 specialist transfer	`evidence/gp216_queries/gp216_sparse_coverage_OOD.json`
PDE correction	`experiments/joint_vocab_corpus_ood_vC01_20260521/`
Catalog reliability stress test	`experiments/v128b_catalog_stats_20260521/`
Catalogue surface tests	`experiments/source_packet_catalog_surface_v14/` and follow-up catalogue/card tests in `research_log.md`
Meta-language tests	Meta-card, typed-frame, and packaging-ablation tests in `research_log.md`
Boundary-card and action-schema tests	Boundary-card and process-control tests in `research_log.md`
Pattern/action contract tests	Pattern/action contract synthesis in `research_log.md` and `research_roadmap.md`
Literature audit	`literature_audit.md`
Claim guardrail	`claim_evidence_matrix.md`

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